



Original Article

A recurrent neural network for modeling natural circulation density wave instabilities

Paul Hurley^{a,*}, Juliana Pacheco Duarte^b^a Department of Mechanical Engineering, Virginia Tech, Blacksburg, VA, USA^b Department of Nuclear Engineering and Engineering Physics University of Wisconsin-Madison, Madison, WI, USA

ARTICLE INFO

Keywords:

Recurrent neural network
Density wave oscillation

ABSTRACT

Advancements in machine learning and deep learning capabilities over several decades have resulted in models specialized for unique data structures and use cases. One such model, the recurrent neural network (RNN), is a deep learning architecture that incorporates temporal memory in its training. This is primarily performed utilizing long short-term memory (LSTM). Compared to dense neural networks (DNN), this feature gives the potential to model highly dynamic behaviors that would otherwise be limited using conventional machine learning models. LSTM is used here to predict two-phase flow instability, specifically density wave oscillation (DWO), which is characterized by significant oscillations in a boiling channel that can severely impact the safety of boiling water reactors (BWRs). Density waves produced by dynamic voiding in the channel dominate the system and maintain high amplitude flow oscillations which can cause cycles of dryout and rewet. In this paper, the RNN model is trained on experimental DWO data in order to test the abilities of the LSTM framework. Training data is obtained from tests using a single channel 10×10 BWR bundle geometry with electrically heated rods at 7–8 MPa. The model is tested with different architectures and utilizes different optimization methodologies to determine the impact on results.

1. Introduction

Density wave oscillation (DWO) is a type of two-phase flow instability characterized by high-amplitude oscillatory behavior developed from unstable void production in the boiling channel [1]. This instability is caused by time delays in the system due to the difference in propagation time between pressure waves and density waves (void propagation) [2,3]. Instabilities of this form are common in conditions with high vapor production such as boiling water reactors (BWRs), and are a source of concern for transient scenarios where flow is reduced to natural circulation [4]. If left unchecked and allowed to develop, the oscillations can grow to the point where dryout of the fuel rod surface can occur at the trough of the flow fluctuation [5]. When the flow rate swings back up, the rod surface can then regain contact with liquid coolant (rewetting). This cycle of dryout and rewet can be severely damaging to the fuel structural integrity. Even worse, if the cycle grows significant enough, the rods can eventually fail to rewet (FTR) [6]. In this event, fuel rod temperature can drastically increase and rod failure is highly likely. Thus, ensuring proper prediction capabilities for this phenomenon is very important to the safety of BWRs.

Developing models to predict the DWO phenomenon can be difficult due to the highly complex nature of the flow instability. Simplified analytical models have been developed using drift flux formulations [7] which have been modified to account for thermal non-equilibrium between the phases [8]. More accurate models have been developed using computational fluid dynamics (CFD) [9], but these can be significantly computationally expensive. Further, they are likely quite far from being useable in real-time simulation. As such, alternative methodologies have been explored to bridge the gap between accuracy and computational efficiency.

Machine learning is a class of artificial intelligence focused on developing predictive models that can improve with successive iterations given a dataset it uses to train itself. Since its onset, numerous machine learning models have been developed for use in an increasingly wide range of tasks from regression and classification to data clustering [10] and dimensionality reduction [11,12]. Physics-informed neural networks have been used in modeling numerous accident scenarios and thermal hydraulic transients [13,14]. Convolutional neural networks (CNN) have been used in instability mode prediction for power systems with good success [15] and hybrid neural networks have been used to

* Corresponding author.

E-mail address: prhurley@vt.edu (P. Hurley).<https://doi.org/10.1016/j.net.2024.103407>

Received 25 July 2024; Received in revised form 7 December 2024; Accepted 22 December 2024

Available online 24 December 2024

1738-5733/© 2024 Korean Nuclear Society, Published by Elsevier Korea LLC. All rights are reserved, including those for text and data mining, AI training, and similar technologies. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

model and forecast wind power [16].

The neural network (also referred to as the multilayer perceptron model) is one of the most well-studied and developed types of machine learning designs used for regression problems [17,18]. The typical architecture of dense neural networks (DNNs) contains a number of input variables and a series of hidden layers of “neurons” (see Fig. 1). Each neuron is composed of a value and a weight W , which is adjusted in order to account for the impact that an input has on the output (in this case the predicted flow stability). The weights are summed using an arbitrary bias b in the form of a hypothesis $h = Wx + b$ [19]. For multiple inputs to the neuron, this is taken as:

$$h_{ij} = \left(\sum_{n=1}^N W_{in} \cdot x_n \right) + b_i \quad (1)$$

The signal from each layer is moved to the next layer following nonlinearization using an activation function. Prediction accuracy is improved during back-propagation using a gradient descent methodology. In addition, certain optimization algorithms can be utilized to further improve accuracy [20].

DNNs are well-used due to their variability and have been developed for a wide variety of tasks. In the engineering sector, DNNs have been used in uncertainty quantification [21,22], prediction of boiling heat transfer [23], and probabilistic safety assessments [24]. Their use in the nuclear sector has been expanding lately across all fields, particularly for use with real-time safety evaluation and monitoring (nuclear power plant monitoring has been implemented with neural networks for abnormality detection [25]).

DNN models are fairly capable of predicting the general behavior of the system under prototypical conditions. Modeling stability maps using DNNs has been performed with some degree of success [26]. However, the model must be trained using averaged data points due to the rapid transient behavior that occurs during instabilities. In order to be able to predict with much higher accuracy, the machine learning model must be able to incorporate time-wise dynamic trends (i.e., the actual oscillations as a function of time). To accomplish this, another class of neural networks must be used: the recurrent neural network (RNN).

RNNs are specialized neural networks that utilize memory keeping in training in order to better model time dependent behavior. These models commonly use a long short-term memory (LSTM) architecture due to its ability to resolve the vanishing error problem found in RNNs (this is

discussed further in the following section). In nuclear systems that are highly non-linear, being able to properly model time dependency can be critical towards improving detection capabilities in rapid transients. For the purposes of this study, DWO-type instability is modeled in order to demonstrate the capability of LSTM in predicting slow and rapid oscillatory behavior. This model is trained and tested using experimental natural circulation data from the KATHY facility.

2. Recurrent neural networks

The RNN model holds many similarities to the DNN in that it incorporates a layered multi-neuron architecture between the input and output nodes. The primary difference is that the RNN also allows for recurrent connections (see Fig. 2) which gives the model the ability to recognize trends between earlier and later inputs temporally. By doing so, the model is able to build high-accuracy predictions of transient phenomena based on the previous outputs.

A further development in RNN architecture is the utilization of long short-term memory (LSTM) [27,28]. This methodology was developed to address the shortcomings of typical RNN models – that when there are more than 5–10 discrete time steps between input variables and relevant output signals, the RNN often fails to properly learn. This problem (known as the vanishing error problem) can be resolved by the LSTM’s use of the so-called “constant error carousel” (CEC, seen in Fig. 3) [29, 30]. In each LSTM layer, there are a number of memory blocks, each containing one or more memory cells C as well as three gates. These gates, called “input”, “output”, and “forget”, act as “write”, “read”, and “reset” operators for the memory cells within the block. The opening and closing of these gates allow for longer memory connections while also affording certain protections for the memory block itself as well as the blocks around it. The “forget” gate is a sigmoid function with x_t and h_{t-1} as inputs:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

This calculates a weight for each cell state C_{t-1} , in essence determining how much of the previous cell state is “remembered”. Next, x_t and h_{t-1} are fed into a tanh function which is used to determine what information from the input should be added to long-term memory:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (3)$$

The input gate is a separate sigmoid function which determines the weight of the data determined by the tanh function to send the memory cell:

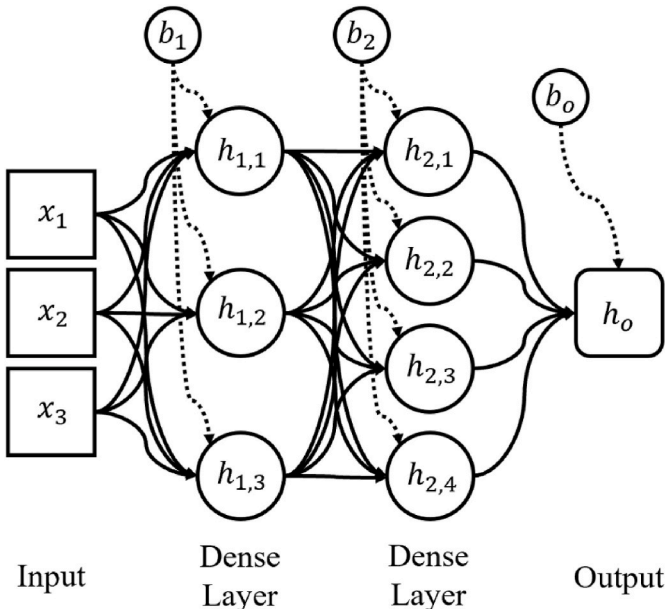


Fig. 1. Layer architecture of DNN [19].

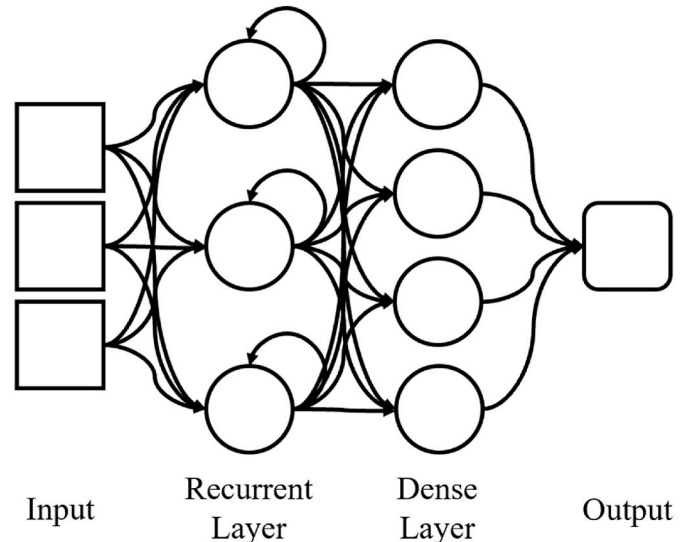


Fig. 2. Modified layer architecture in RNN.

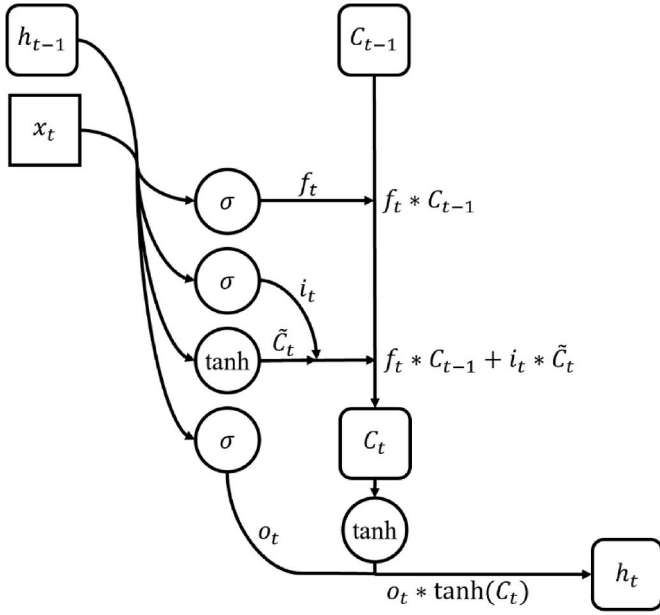


Fig. 3. Constant error carousel structure.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (4)$$

This allows the input gate to minimize perturbations from irrelevant inputs. By combining the data from these first two gates, the current cell state can be calculated:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (5)$$

The output gate is a final sigmoid function on x_t and h_{t-1} :

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

The current cell state is fed into another tanh function to determine what to potentially add to h_t , which is then multiplied by o_t :

$$h_t = o_t * \tanh(C_t) \quad (7)$$

Similar to the other gates, o_t is used to weight the data being stored in h_t . This way, the output gate protects other blocks from being perturbed by currently irrelevant information stored in this block.

Further variations on LSTM architectures have been developed for specific use cases. So-called “peephole” LSTM models, where each gate is given the previous cell state C_{t-1} as another input, has been used to recognize precisely timed events even over larger time spans [31]. Another variation, the Gated Recurrent Unit (GRU), combines the “forget” and “input” gates into a single gate, as well as combining the cell states and hidden states [32]. This is a more simplistic version of the LSTM that has shown similar performance to standard LSTM in certain tasks such as speech recognition [33]. Notably, GRU-based models have been tested for sensor fault detection and accident diagnosis, finding that there was an inherent time delay in detecting faults that could be improved on via changes to the model itself and to the decision logic implemented [34].

RNN models in general have been used previously in a wide range of applications such as speech recognition [35,33], audio analysis [36], stock market analysis [37], and CHF prediction [38], indicating their strengths in a range of transient datasets. Kim et al. [39] utilized RNN for anomaly detection by training from multi-sensor signals including anomalous and anomaly-free data. In this study, the authors found that incorporating uncertainties in anomaly times was beneficial to training the capable model, particularly when there is a lack of existing anomalous data to train on. While anomaly detection may have limited training data, instability data can be more easily obtained that contains both stable and oscillatory flow patterns. This is the primary reason for

assessing the RNN, and particularly the LSTM, in predicting oscillatory behavior. In the following section, an LSTM-based RNN model is trained on data from a series of flow instability experiments from the Karlstein thermal hydraulic test facility (KATHY) [40] in order to assess its capability of modeling oscillatory conditions. To do this, the model is trained to predict the inlet mass flow rate given other measurable or controllable system conditions such as subcooling and power. A properly-trained model with temporal memory can prove exceedingly beneficial to reactor safety, as rapid decision-making is necessary during DWO to prevent damage to fuel integrity. By showcasing the strengths of this model in predicting flow oscillation, LSTMs can be advantageous in other areas of design and anomaly detection.

3. Model training dataset

The dataset used for training these models is provided by the KATHY facility’s series of density wave instability experiments. The Karlstein thermal hydraulic test facility is used primarily for critical heat flux (CHF) testing for both pressurized water reactors (PWRs) as well as BWRs. Components, including the boiling channel, are configurable to test a variety of thermal hydraulic transients. A natural circulation BWR test loop is used for the experiments this dataset is pulled from. Fig. 4 shows the layout of the facility for these tests. Natural circulation flow is maintained via a large diameter downcomer with supplied feedwater flow to supplement lost coolant mass due to boiling. A steam separator removes vapor at the outlet and re-introduces liquid to the loop. The boiling channel consists of a 10×10 BWR rod bundle (comprised of 81 full-length and 10 part-length rods) with spacer grids to assist in flow orientation control and a water channel. The electrically heated rods are designed to produce a bottom-peaked axial power profile. Additional non-proprietary information is provided in NUREG/CR-7272 [40].

Inlet flow during steady state or slow transients is measured with a turbine flow meter. For oscillatory conditions, flow is measured based on the inlet pressure drop:

$$I \frac{d\dot{m}}{dt} = \Delta P(t) - \frac{k}{2\rho A^2} \dot{m}(t) |\dot{m}(t)| \quad (8)$$

Here, I is the flow inertia, $\dot{m}$ is the mass flow rate, ΔP is the pressure difference, k is the loss coefficient, ρ is the density, and A is a reference area used to ensure accuracy in steady state and transient conditions.

For this study, the experiments analyzed were conducted by incrementally increasing power to the rod bundle and observing flow behavior. When flow instability causes oscillations to grow to a certain level, the power is quickly scaled down in order to halt the transient. Tests were performed at 20 and 45 °C inlet subcooling. In certain experiments, the oscillation was allowed to continue and grow until the dryout/rewet cycle and FTR was reached (determined as a surface temperature excursion not followed by rewet for at least two mass flow oscillations). Table 1 shows the relevant test matrix of the experimental dataset, including measurement uncertainties for each parameter. Hydraulic data such as flow rate and density are taken at the channel inlet. Since the total power capabilities of this facility are kept proprietary, only the relative power is shown in this table. Each input variable was chosen for its noted impact on two-phase instability.

Table 2 shows the inlet parameters for each experimental condition. Here, N_{sub} is the subcooling number defined in Eq. (9) and $\dot{m}_{avg}$ is the average inlet mass flow rate. The training dataset comprises a total of approximately 154,000 data points taken from 13 tests. Fig. 5 shows flow rate data from each experiment. Cases consist of stable, marginally stable, and unstable operating conditions as given by flow oscillation behavior. This allows the model to be trained and tested on flow behavior that varies significantly. Additionally, several cases contain flow data during FTR along with rapid shutdown which can be used to test rapid transients. Data are taken at a frequency of 20 Hz.

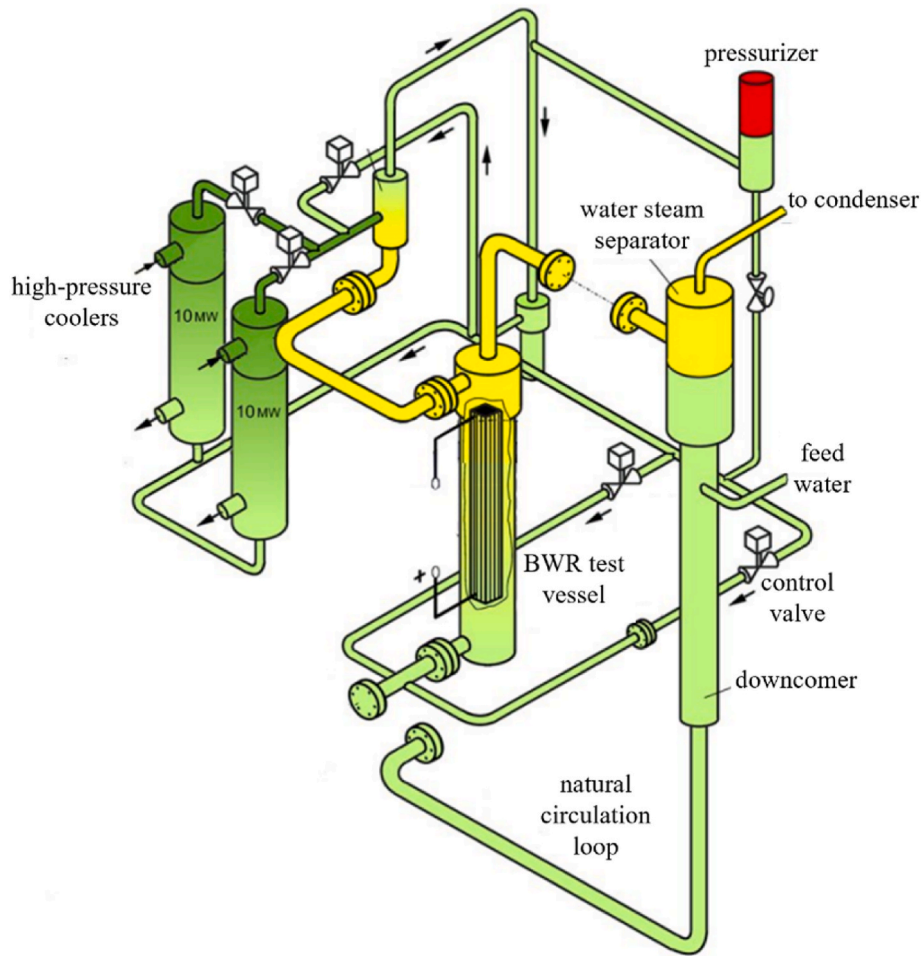


Fig. 4. KATHY thermal hydraulic test facility.

Table 1

Test matrix for input data.

Input Variable	Range	Uncertainty
Flow Rate [kg/s]	3.2–8.6	±1.10 %
Relative Power [–]	0.76–1.18	±0.20 %
Vapor Density [kg/m ³]	36.3–42.2	±0.14 %
Liquid Density [kg/m ³]	722.3–740.2	±1.16 %
Inlet Subcooling [kJ/kg]	48–124	±5.83 %

Table 2

Inlet parameters across all experiments [40].

Experiment	P [MPa]	$\dot{m}_{avg}$ [kg/s]	N _{sub}
105.02	7.02	4.70	1.42
106.01	7.00	4.63	1.36
107.01	6.98	4.55	1.42
108.01	6.98	4.45	1.37
109.01	7.05	4.35	1.37
110.01	6.98	4.10	1.38
205.01	6.98	4.25	1.34
206.01	6.99	4.00	1.37
207.01	7.01	4.00	1.36
207.02	7.05	4.00	1.35
207.03	7.01	4.00	1.33
406.01	7.88	5.00	1.19
407.01	7.94	4.35	1.15

$$N_{sub} = \frac{\Delta i_{sub}}{\Delta i_{fg}} \frac{\Delta \rho}{\rho_g} \quad (9)$$

The RNN model is trained on the system power as well as the densities and enthalpies of the coolant at the channel inlet, which are functions of the inlet pressure and temperature. The TensorFlow library is used to train the model, which is built using the LSTM and Dense classes from the Keras 3 API, used for developing neural networks using Python. The output feature of the model is the inlet flow rate, which can be directly compared to the test measurements. Training and testing data was split in an 80:20 ratio, respectively. In some cases, such as experiment 407.01, this results in the model being trained on stable or low-amplitude oscillations, then tested on growing oscillations. This variation allows for better understanding of model performance under different conditions. While the use of LSTM is expected to diminish the impact of vanishing gradient, the dataset is still normalized using the min-max scaling method and limited to a range of 0.1–0.9 in order to reduce biasing from outliers (Eq. (10)). For general model testing, the Nadam optimizer (a modification of Adam with Nesterov momentum [41]) is used with an LSTM(50-50)-25 layer architecture (two layers of 50 neurons with CEC along with a single layer of 25 typical dense neurons). The following section will discuss the results of this testing for steady and variable oscillation conditions.

$$X_n = 0.8 \frac{X - X_{min}}{X_{max} - X_{min}} + 0.1 \quad (10)$$

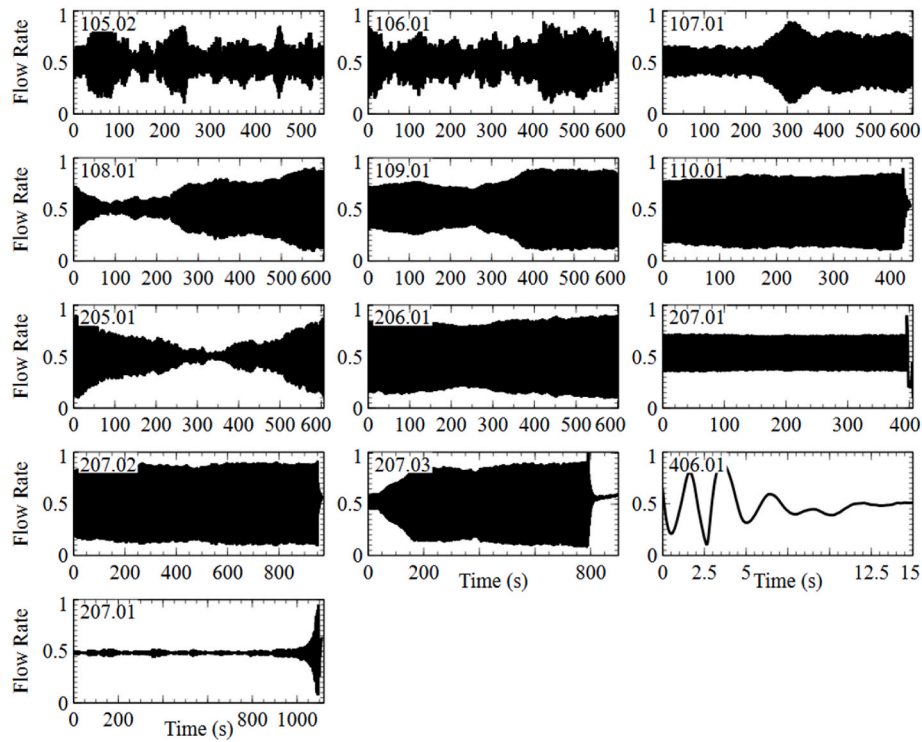


Fig. 5. Mass flow rates across all experiments.

4. Prediction results

The first model to be tested is validated against fairly consistent oscillatory conditions (in Fig. 5, from experiment 207.03). The intent behind this test is to ensure that the general model is trained correctly and can accurately reproduce the flow behavior when given data that only slightly vary outside of the main oscillation frequency and amplitude. As shown in Fig. 6, the RNN model allows for highly accurate predictions within the testing region in tracking the flow amplitude, frequency, and shape of the oscillation under consistent conditions.

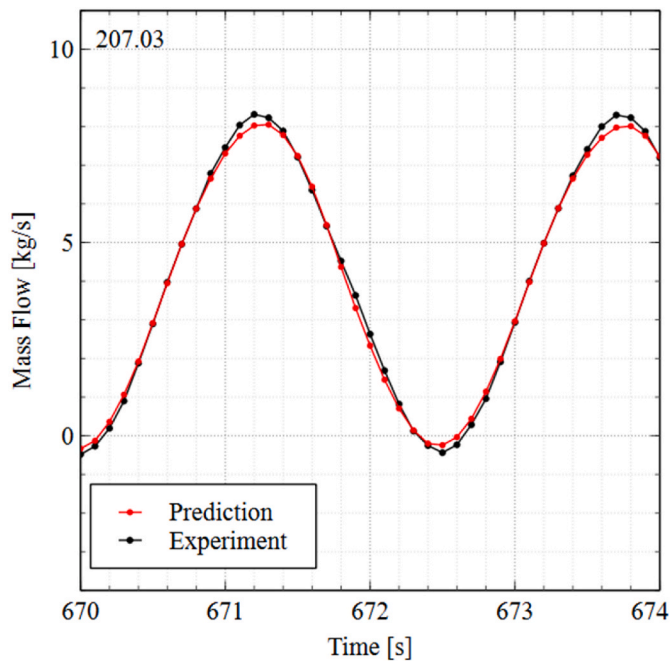


Fig. 6. Mass flow prediction under consistent oscillatory conditions.

The next model is tested on data containing oscillations that vary in their behavior. For the purposes of this study, data from experiment 407.01 are shown since it contains both slower changes (a smoothly exponential amplitude increase) as well as rapid changes (oscillation breakdown following a sudden drop in power at the end of the experiment). This is intended as a stress test for the RNN model, as the dryout/rewet cycle and FTR event provide additional complexity to the system that typical DWO behavior does not have. Fig. 7 shows model performance for this test. In addition, Fig. 8 compares RNN model

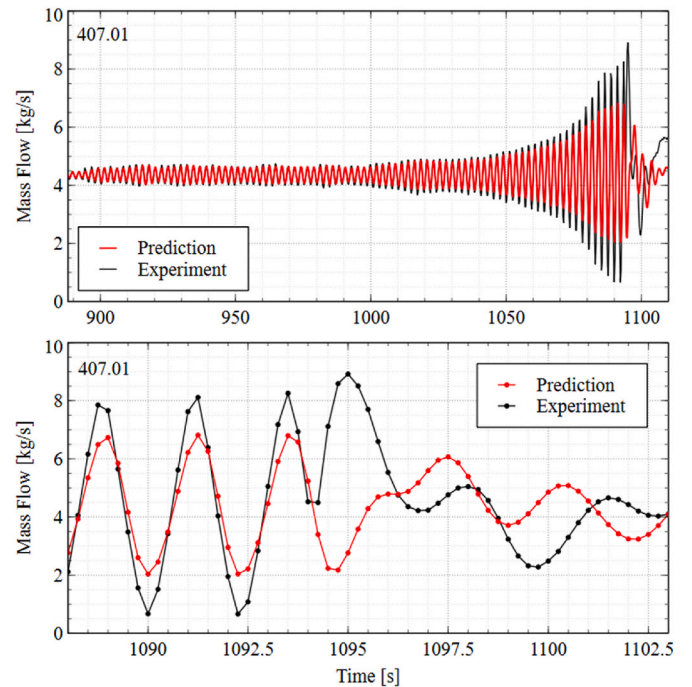


Fig. 7. Mass flow prediction under variable oscillatory conditions.

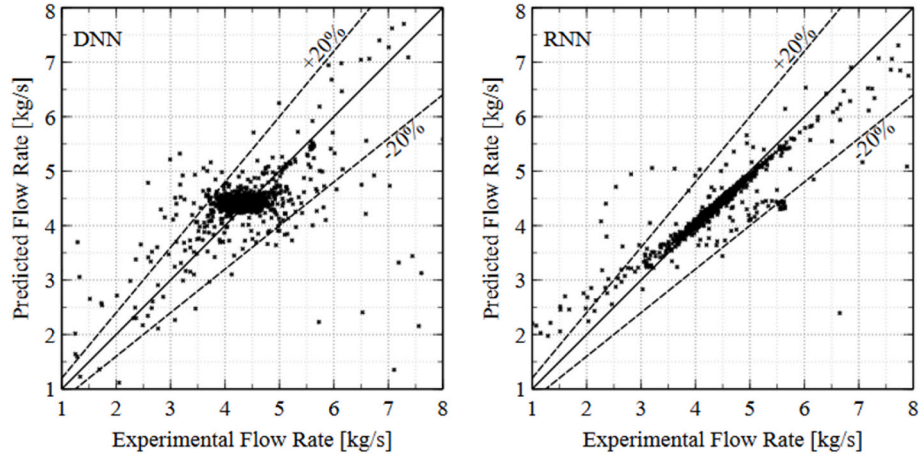


Fig. 8. Comparison of RNN performance to typical DNN for test 407.01.

performance against a typical DNN model that was trained on the same dataset. In this case, the DNN architecture was comprised of three dense layers of 50 neurons each and also used the Nadam optimizer. As shown, the DNN has difficulty matching the flow rate due to the strong variation during oscillatory conditions. While it is possible to use DNN to predict general instability onset (and this has in fact been done in a prior study [26]), it falls short of being useable in detecting changes in flow behavior dynamically. The RNN, by comparison, is much more capable at tracking the flow behavior over time, resulting in a much more accurate model.

Results are compared across both tests in Table 3. Total experiment duration is the sum of training and testing time (e.g. experiment 207.03 was 903 s split into 722 s for training and 181 s for testing). Performance is rated via three main factors: a comparison of the predicted average flow amplitude A_{pred} versus the actual measured amplitude A_{exp} (calculated as a percentage of the actual amplitude), the root mean square error (RMSE, Eq. (11)), and the R^2 score (Eq. (12)). The RMSE is indicative of the dispersion between RNN predicted values and experimentally measured values, and R^2 is the coefficient of determination. Higher accuracy models will show lower RMSE and R^2 values closer to 1 indicate that the output value is highly dependent on the input.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{exp,i} - y_{pred,i})^2} \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_{exp,i} - y_{pred,i})^2}{\sum_{i=1}^N (y_{exp,i} - \bar{y}_{exp,i})^2} \quad (12)$$

Overall prediction accuracy is high even with variable oscillation testing. The most notable difference in performance is with the flow amplitude prediction. As shown, the RNN model under-predicts the flow amplitude with increased severity in the variable oscillation test, with the worst performance at the very end of the experiment before power is shut down. Fig. 9 shows that there is increased error with higher amplitude flow (note that the error is adjusted for amplitude and still shows this trend). While the oscillations are predicted with accurate

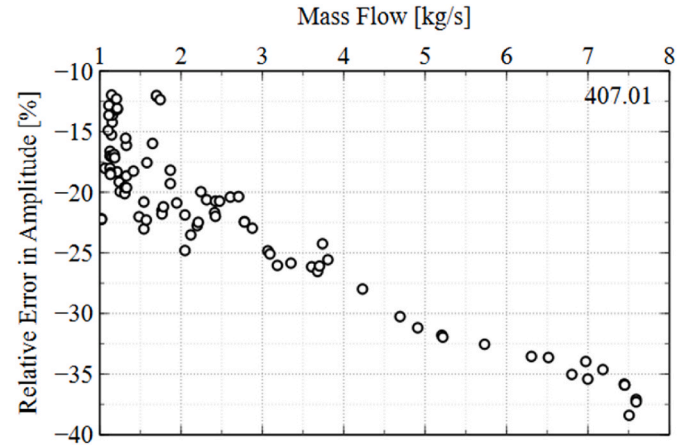


Fig. 9. Error in predicted amplitude versus mass flow rate for experiment 407.01.

periodic behavior since it is a function of the loop geometry, the amplitude appears dampened. This would be considered a non-conservative estimate in traditional models as higher amplitude oscillations provide greater risk to the system. This underprediction is potentially due to sub-optimal training hyperparameters, of which alternatives are discussed in the following section. In addition, the RNN as it is currently built is not capable of accurately predicting the flow behavior upon power shutdown, primarily the flow peaking upon reaching FTR. This is expected, as the model is trained on limited data for this transient. Still, it is interesting to note that the general behavior (continued oscillation with rapidly decreasing amplitude) is still in line with the experimental measurements. Still, the model can be further optimized by adjusting the model layer architecture and testing different optimizer performances. This parametric study is performed in the following section.

5. Parametric investigation

RNN performance can be significantly affected by the parameters used for training the model. In this section, the impact of optimizers and layer architecture on modeling accuracy is demonstrated.

The Keras API allows for a variety of optimizers to be implemented in model training. Additionally, different layer architectures can be configured to improve fitting performance and balance training and testing RMSE. In the case of this study, the default parameters for each optimizer (such as learning rate) are used. A total of 24 configurations

Table 3
Prediction results for steady and variable oscillation conditions.

Experiment	Training/Test Duration [s]	A_{pred}/A_{exp} [%]	Test RMSE	Test R^2
207.03 (Steady)	722/181	98.7	0.0166	0.9641
407.01 (Variable)	888/222	86.2	0.0612	0.8351

are tested for this study, comparing four commonly used optimizers and six layer architectures. The chosen optimizers, Adam, Nadam, RMSprop, and stochastic gradient descent (SGD), are used as they are most commonly applicable for these kinds of models. Table 4 through 6 show the training and testing performance for each configuration for experiment 407.01. Ideal performance shows comparably (low) training and testing RMSE, indicating that no overfitting is occurring, and R^2 values close to 1. From the configurations tested, the optimal model has an LSTM(150-150)-25 layer architecture and uses the Nadam optimizer. This configuration showed solid testing RMSE performance without overfitting (as indicated by high test R^2). Fig. 10 shows the prediction performance of this configuration. Average predicted flow amplitude in tests matched measured amplitude to 97.1 %. Notably, this model shows that it no longer underpredicts the oscillation amplitude, providing a conservative estimate on the flow behavior. Other high-performing configurations include LSTM(25-25)-25 with RMSprop and LSTM(200-200)-25 with Adam. The SGD optimizer showed generally poor performance overall. Table 7 shows the prediction performance over each test using the optimized model architecture. As can be seen with experiment 406.01, model performance can be limited when trained on short time spans. However, all other test cases show very good prediction performance.

6. Conclusions

An LSTM-based recurrent neural network was trained on experimental density wave oscillation data to determine its performance in modeling flow oscillations and instability boundaries. The DWO dataset was pulled from the KATHY facility using a 10×10 BWR rod bundle geometry under natural circulation at 7–8 MPa. The trained model was tested on separate untrained data for test cases with steady and variable oscillation conditions. The general model showed very good prediction results. Specifically, the trained model was able to accurately match the oscillation period and general amplitude growth. Performance was best with steady oscillation conditions, where predicted amplitudes matched experimental measurements above 98 %. Variable oscillation conditions showed worse performance (primarily following simulated scram), particularly underpredicting the oscillation amplitude. However, adjusting the model hyperparameters and finding the ideal optimizer resulted in flow matching measured amplitudes with above 97 % accuracy and eliminating this underprediction.

Overall, the LSTM-based RNN showed excellent performance in modeling and predicting DWO-type instability. It is notable that this high performance is possible even without the influence of physical models or experimental correlations. It is expected that with the inclusion of physics-based methodologies such as those used in other studies [13,14] the performance would improve even further, particularly under more rapid transients such as during the reactor scram. Further study with variations on the typical LSTM architecture such as those mentioned in Section 2 are recommended for better insight on improvements to prediction capabilities. The expansion of LSTM-based RNN models is also being explored to predict DWO with neutronic feedback response to more closely model the behavior inside a BWR. Additionally, this model is expected to provide strong results outside of DWO-type transients, allowing for a wide range of potential use cases

Table 4
Training RMSE performance for experiment 407.01

Architecture	Adam	Nadam	RMSprop	SGD
LSTM(25-25)-25	0.0029	0.0045	0.0056	0.0028
LSTM(50-50)-25	0.0122	0.0023	0.0047	0.0016
LSTM(75-75)-25	0.0017	0.0038	0.0012	0.0004
LSTM(100-100)-25	0.0015	0.0006	0.0057	0.0018
LSTM(150-150)-25	0.0023	0.0017	0.0032	0.001
LSTM(200-200)-25	0.0027	0.0012	0.0047	0.0055

Table 5
Test RMSE performance for experiment 407.01

Architecture	Adam	Nadam	RMSprop	SGD
LSTM(25-25)-25	0.0678	0.0681	0.0097	0.0679
LSTM(50-50)-25	0.0664	0.0612	0.0208	0.0671
LSTM(75-75)-25	0.0197	0.0129	0.0165	0.0677
LSTM(100-100)-25	0.0117	0.0106	0.0119	0.0662
LSTM(150-150)-25	0.0216	0.0107	0.016	0.0683
LSTM(200-200)-25	0.0095	0.0149	0.016	0.0679

Table 6
Test R^2 performance for experiment 407.01

Architecture	Adam	Nadam	RMSprop	SGD
LSTM(25-25)-25	0.1839	0.5254	0.9854	0.4765
LSTM(50-50)-25	0.2718	0.8351	0.9322	0.4611
LSTM(75-75)-25	0.9848	0.9682	0.9659	0.4002
LSTM(100-100)-25	0.9823	0.9895	0.9687	0.4651
LSTM(150-150)-25	0.9528	0.9914	0.9430	0.5575
LSTM(200-200)-25	0.9843	0.9648	0.9858	0.7650

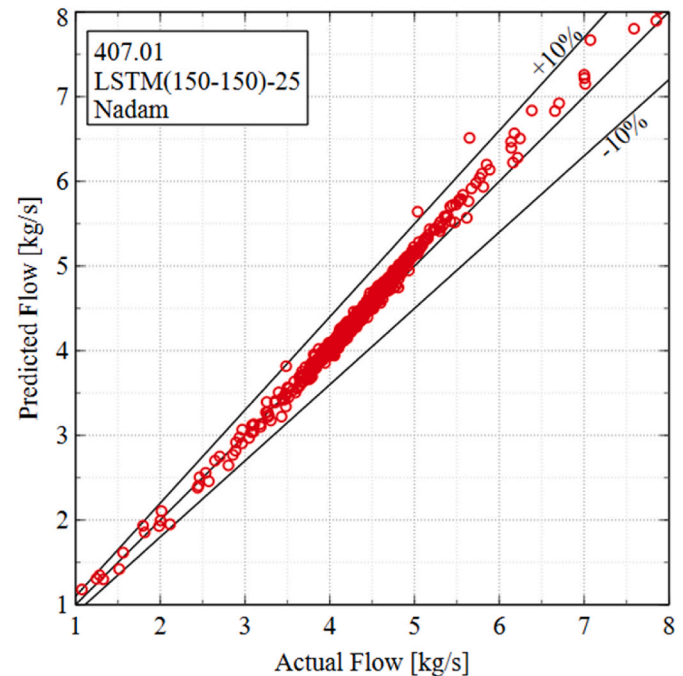


Fig. 10. Prediction performance with optimized RNN model.

Table 7
Prediction performance with optimized LSTM model across all experiments.

Experiment	Training/Test Duration [s]	Training RMSE	Test RMSE	Test R^2
105.02	451/113	0.0008	0.0175	0.9722
106.01	486/122	0.0165	0.0246	0.9875
107.01	482/121	0.0046	0.0073	0.9991
108.01	482/120	0.0034	0.0099	0.9990
109.01	486/122	0.0020	0.0067	0.9995
110.01	348/87	0.0052	0.0088	0.9993
205.01	485/121	0.0125	0.0174	0.9958
206.01	482/120	0.0065	0.0142	0.9998
207.01	325/81	0.0090	0.0420	0.9423
207.02	773/193	0.0116	0.0169	0.9998
207.03	722/181	0.0008	0.0066	0.9998
406.01	12/3	0.0131	0.0141	0.7382
407.01	888/222	0.0017	0.0107	0.9914

within the nuclear field. Such design basis cases as loss of coolant accidents (LOCA) and reactivity insertion accidents (RIA) are suggested for future research.

CRedit authorship contribution statement

Paul Hurley: Writing – original draft, Formal analysis, Conceptualization. **Juliana Pacheco Duarte:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the U.S. Nuclear Regulatory Commission under award 31310019M0015.

References

- [1] L.A. Belblidia, C. Bratianu, Density-wave oscillations, *Ann. Nucl. Energy* 6 (1979) 425–444.
- [2] G. Yadigaroglu, A.E. Bergles, An Experimental and Theoretical Study of Density Wave Oscillations in Two-phase Flow, DSR vols. 74629–3, Massachusetts Institute of Technology, 1969.
- [3] J. March-Leuba, Density-wave Instabilities in Boiling Water Reactors. NUREG/CR-6003, ORNL/TM-12130, Oak Ridge National Laboratory, 1992.
- [4] D.D.B. van Bragt, T.H.J.J. van der Hagen, Stability of natural circulation boiling water reactors: part i - description stability model and theoretical analysis in terms of dimensionless groups, *Nucl. Technol.* 121 (1998) 40–51.
- [5] J.P. Duarte, P. Yarsky, T. Zaki, M.L. Corradini, Full-scale dryout/rewet instability tests in a BWR rod bundle and TRACE assessment, *Nucl. Eng. Des.* 370 (2020) 110930.
- [6] L.-Y. Cheng, J.S. Baek, A. Cuadra, A. Aronson, D. Diamond, P. Yarsky, TRACE/PARCS analysis of anticipated transient without scram with instability for a MELLRA+ BWR/5, *Nucl. Technol.* 196 (2017) 238–247.
- [7] M. Ishii, Thermally Induced Flow Instabilities in Two-phase Mixtures in Thermal Equilibrium, Georgia Institute of Technology, 1971. PhD thesis.
- [8] P. Saha, N. Zuber, An analytical study of the thermally induced twophase flow instabilities including the effect of thermal non-equilibrium, *Int. J. Heat Mass Tran.* 21 (1978) 415–426.
- [9] S.L. Sharma, J.R. Buchanan, M.A. Lopez de Bertodano, Density wave instability verification of CFD two-fluid model, *Nucl. Sci. Technol.* 194 (2020) 665–675.
- [10] K.-L. Du, Clustering: a neural network approach, *Neural Network.* 23 (2010) 89–107.
- [11] A. George, Anomaly detection using machine learning: dimensionality reduction using PCA and classification using SVM, *Int. J. Comput. Appl.* 47 (21) (2012) 5–8.
- [12] G. Thippa Reddy, M. Praveen Kumar Reddy, Kuruva Lakshmana, Rajesh Kaluri, Dharmendra Singh Rajput, Gautam Srivastava, Thar Baker, Analysis of dimensionality reduction techniques on big data, *IEEE Access* 8 (2020) 54776–54788.
- [13] K.M. Kim, P. Hurley, J.P. Duarte, Physics-informed machine learning-aided framework for prediction of minimum film boiling temperature, *Int. J. Heat Mass Tran.* 191 (2022) 122839.
- [14] F. Antonello, J. Buongiorno, E. Zio, Physics informed neural networks for surrogate modeling of accidental scenarios in nuclear power plants, *Nucl. Eng. Technol.* 55 (9) (2023) 3409–3416.
- [15] Z. Shi, W. Yao, L. Zeng, J. Wen, J. Fan, X. Ai, J. Wen, Convolutional neural network-based power system transient stability assessment and instability mode prediction, *Appl. Energy* 263 (2020) 114586.
- [16] Y.-Y. Hong, C.L.P.P. Rioflorida, A hybrid deep learning-based neural network for 24-h ahead wind power forecasting, *Applied Power* 250 (2019) 530–539.
- [17] K. Hornik, M. Stinchcombe, H. White, Multilayer feedforward networks are universal approximators, *Neural Network.* 2 (5) (1989) 359–366.
- [18] J. Thibault, B.P.A. Grandjean, A neural network methodology for heat transfer data analysis, *Int. J. Heat Mass Tran.* 34 (8) (1991) 2063–2070.
- [19] K.M. Kim, P. Hurley, J.P. Duarte, High-resolution prediction of quenching heat behavior using machine learning based on optical fiber temperature measurement, *Int. J. Heat Mass Tran.* 184 (2022) 122338.
- [20] D.P. Kingma, J. L. Ba Adam, A method for stochastic optimization, in: ICLR 2015 - International Conference on Learning Representations, 2015.
- [21] R.K. Tripathy, I. Bilionis, Deep UQ: learning deep neural network surrogate models for high dimensional uncertainty quantification, *J. Comput. Phys.* 375 (2018) 565–588.
- [22] E. Sahin, B. Lattimer, M.A. Allaf, J.P. Duarte, Uncertainty quantification of unconfined spill fire data by coupling Monte Carlo and artificial neural networks, *J. Nucl. Sci. Technol.* 61 (9) (2024) 1218–1231.
- [23] G. Scalabrini, M. Condosta, P. Marchi, Modeling flow boiling heat transfer of pure fluids through artificial neural networks, *Int. J. Therm. Sci.* 45 (7) (2006) 643–663.
- [24] C. Worrell, L. Luangkesorn, J. Haight, T. Congedo, Machine learning of fire hazard model simulations for use in probabilistic safety assessments at nuclear power plants, *Reliab. Eng. Syst. Saf.* 183 (2019) 128–142.
- [25] K. Nabeshima, T. Suzudo, K. Suzuki, E. Turkan, Real-time nuclear power plant monitoring with neural network, *J. Nucl. Sci. Technol.* 35 (2) (1998) 93–100.
- [26] P. Hurley, J.P. Duarte, Prediction of unstable two-phase flow behavior using dense neural network, in: 2022 ANS Winter Meeting and Technology Expo, 2022.
- [27] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput.* 9 (8) (1997) 1735–1780.
- [28] Y. Yu, X. Si, C. Hu, J. Zhang, A review of recurrent neural networks: LSTM cells and network architectures, *Neural Comput.* 31 (2019) 1235–1270.
- [29] S. Yao, L. Lao, H. Peng, High-frequency stock trend forecast using LSTM model, in: 13th International Conference on Computer Science & Education (ICCSE), 2018.
- [30] P. Ramaraj, A neural network in convolution with constant error carousel based long short term memory for better face recognition, *Turkish J. Comp. Math. Edu.* 12 (2) (2021) 2042–2052.
- [31] F.A. Gers, N.N. Schraudolph, J. Schmidhuber, Learning precise timing with LSTM recurrent networks, *J. Machine Learning* 3 (2002) 115–143.
- [32] K. Cho, B. van Merriënboer, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using RNN encoder-decoder for statistical machine translation. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2014.
- [33] M. Ravanelli, P. Brakel, O. Philemon, M. Omologo, Y. Bengio, Light gated recurrent units for speech recognition, *IEEE Transac. Emerging Topics in Computational Intelligence* 2 (2) (2018) 92–102.
- [34] J. Choi, S.J. Lee, RNN-based integrated system for real-time sensor fault detection and fault-informed accident diagnosis in nuclear power plant accidents, *Nucl. Eng. Technol.* 55 (2023) 814–826.
- [35] W. Feng, N. Guan, Y. Li, X. Zhang, Z. Luo, Audio visual recognition with multimodal recurrent neural networks, in: 2017 International Joint Conference on Neural Networks (IJCNN), 2017.
- [36] N. Boulanger-Lewandowski, Y. Bengio, P. Vincent, Audio chord recognition with recurrent neural networks, in: International Society for Music Information Retrieval Conference, 2013.
- [37] K. Kamijo, T. Tanigawa, Stock price pattern recognition - a recurrent neural network approach, in: 1990 International Joint Conference on Neural Networks (IJCNN), 1990.
- [38] K. Marcinkiewicz, O. Wieckhorst, R. Macian-Juan, M. Rehm, P. Polh, Recurrent neural network-based prediction of critical heat flux in rod bundles with non-uniform axial power shape, *Nucl. Eng. Des.* 395 (2022) 111825.
- [39] M. Kim, E. Ou, P. Loh, T. Allen, R. Agasie, K. Liu, RNN-based online anomaly detection in nuclear reactors for highly imbalanced datasets with uncertainty, *Nucl. Eng. Des.* 364 (2020) 110699.
- [40] D. Tinkler, K. Greene, K. Quick, J. Kronenberg, R. Velton, F. Wehle, A. Beisiegel, T. Berger, F. El Rharbaoui, P. Yarsky, NRC ATWS-I Stability Tests with Downsized Axial Power Profile. NUREG/CR-7272, U.S. Nuclear Regulatory Commission, 2020.
- [41] T. Dozat, Incorporating Nesterov momentum into Adam. In the International Conference on Learning Representations, 2016.